Introduction To Flight Mechanics & Control

Tutorial 5



source: flugrevue.de

Important units in aerospace engineering

Knots <-> meter/second

1 m/s = 1.944 knots

or

1 knots = 0.514 m/s

Feet <-> meter

1 m = 3.281 ft

or

1 ft = 0.305 m

Knowing these conversions **could** be helpful in the exam!

Key Facts

A **turbojet aircraft** (symmetric quadratic drag polar) begins to **glide** at a height above ground of h_0=1000 m.

Drag coefficient	C_{D0}	0.031
Drag factor	k	0.026
Wing reference area	S_{ref}	18 m^2
Aircraft mass	m	2000 kg
Maximum engine performance parameter	L_P	6000

Altitude and density	h / ρ	1000 m / 1.1117 $\frac{kg}{m^3}$

Key Facts ISA

MSL density ISA standard atmosphere: $\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$

MSL pressure ISA standard atmosphere: $p_{s,MSL} = 1.01325e5\ Pa$

MSL temperature ISA standard atmosphere: $T_{s,MSL} = 288.15\ K$

Gravitational acceleration: $g_s = 9.80665 \frac{m}{s^2}$

Speed of sound MSL: $a_{s,MSL} = 340 \frac{m}{s}$

Specific gas constant: $R = 287.05 \frac{J}{kg\ K}$

Polytropic exponent: $n = 1.235$

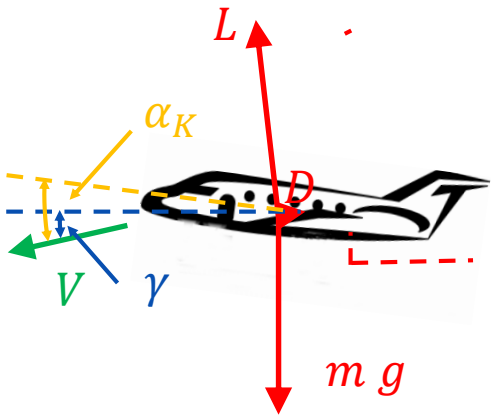
$\kappa = 1{,}4$

Hint: If you know the **PM-EOM** and the **conditions of each flight phase**, you can easily derive all formulas!!

Definition of Gliding:

Straight flight without thrust but constant velocity.

Straight Flight:	$\dot{\chi} = 0$	
Stationary:	$\dot{\gamma} = 0$	$\dot{V} = 0$
No Thrust:	$T = 0$	



source: marinebuzz.com

Parameters of Interest:

Gliding is effective if for a lost meter of altitude a high horizontal distance can be covered (small glide number).

1.	*Glide Number for most effective flight?*	→	ε^*
2.	*Most efficient velocity for gliding?*	→	V^*
3.	*Coefficient of lift for most efficient gliding?*	→	C_L^*
4.	*Coefficient of drag for most efficient gliding?*	→	C_D^*

Equations of Motion and Glide Number

We use the conditions for gliding in the equations of motion:

Hint: If you know the **PM-EOM** and the **conditions of each flight phase**, you can easily derive all formulas!!

$$\dot{V} = \frac{T - D}{m} - g \sin\gamma \overset{!}{=} 0 \quad (T = 0) \longrightarrow D = -m\, g \sin\gamma$$

⇦ Steady state and without thrust

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \cos\gamma} \overset{!}{=} 0 \longrightarrow \mu = 0$$

⇦ Steady state and straight flight

$$\dot{\gamma} = \frac{L \cos\mu}{m\, V} - \frac{g}{V}\cos\gamma \overset{!}{=} 0 \longrightarrow L = m\, g \cos\gamma$$

⇦ Steady state and straight flight

Ratio of drag and lift

$$\frac{-D}{L} = \frac{m\, g \sin\gamma}{m\, g \cos\gamma} = \tan\gamma = -\frac{\frac{1}{2}\rho V^2 S C_D}{\frac{1}{2}\rho V^2 S C_L} = -\frac{C_D}{C_L}$$

$$\varepsilon = \frac{C_D}{C_L} = -\tan\gamma$$

for gliding

The ratio of drag and lift is called **Glide Number ε.** It does not depend on the aircraft mass (or wing loading $\frac{m\,g}{S}$) and altitude ($\widehat{=}\ \rho$). It only depends on the coefficients C_D and C_L. For gliding the glide number equals $-\tan\gamma$.

Calculation of the Performance Parameters V^*, C_L^*, C_D^* and ε^*:

From Equations of Motion $$L = m\,g\underbrace{\cos\gamma}_{\approx 1} = \frac{1}{2}\,\rho\,V^2\,S\,C_L$$

$\cos\gamma \approx 1 - \frac{1}{2}\gamma^2 + \frac{1}{24}\gamma^4 - \cdots$
$\Rightarrow \cos\gamma \approx 1$ for small γ

→ Velocity in gliding: $$V = \sqrt{2}\cdot\underbrace{\sqrt{\frac{m\,g}{S}}}_{\substack{wing\\loading}}\cdot\underbrace{\sqrt{\frac{1}{\rho}}}_{altitude}\cdot\underbrace{\sqrt{\frac{1}{C_L}}}_{aerodynamics}$$

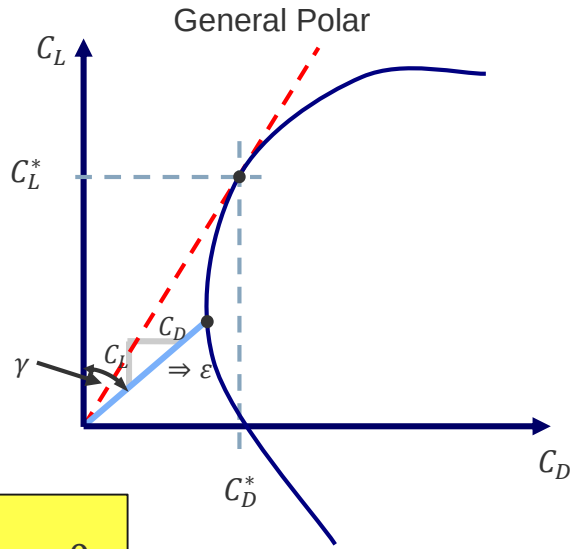
$V \sim \sqrt{\frac{m\,g}{S}}$, $V \sim \sqrt{\frac{1}{\rho}}$
Velocity of gliding depends on wing load and altitude

Idea of Calculations :

Gliding is most efficient, when the ratio $\frac{C_D}{C_L}$ is at minimum, meaning that the glide number and glide angle are also at minimum:

Assumption:
Symmetric quadratic Polar:

$$\varepsilon = \frac{C_D}{C_L} = \frac{C_{D0} + k\,C_L^2}{C_L} = \frac{C_{D0}}{C_L} + kC_L$$



Condition for the minimum:

$$\frac{\partial\varepsilon}{\partial C_L} = 0 \quad \Rightarrow -\frac{C_{D0}}{C_L^2} + k = 0$$

$$\Rightarrow -\frac{C_{D0}}{C_L^2} + k = 0$$

The C_L solving these equations is C_L^*

$$\Rightarrow -\frac{C_{D0}}{(C_L^*)^2} + k = 0$$

Solving for C_L^* and inserting in the equation yields:

$$\frac{C_{D0}}{(C_L^*)^2} = k \quad \xrightarrow{\text{Solving for } C_L^*} \quad C_L^* = \sqrt{\frac{C_{D0}}{k}}$$

$$C_D^* = C_{D0} + kC_L^{*2} \quad \xrightarrow{C_L^* \text{ inserted}} \quad C_D^* = 2C_{D0}$$

$$V = \sqrt{2\frac{m\,g \cdot 1}{S\,\rho\,C_L}} \quad \xrightarrow{C_L^* \text{ inserted}} \quad V^* = \sqrt{2\frac{m\,g \cdot 1}{S\,\rho\sqrt{\frac{C_{D0}}{k}}}}$$

$$\varepsilon = \frac{C_D}{C_L} \quad \xrightarrow{C_L^* \text{ and } C_D^* \text{inserted}} \quad \varepsilon^* = 2\sqrt{k\,C_{D0}}$$

Velocity Polar

The velocity can be split into a horizontal and a vertical part:

$$V_H = \dot{x} = V \cdot \cos\gamma \qquad V_V = \dot{h} = V \cdot \sin\gamma$$

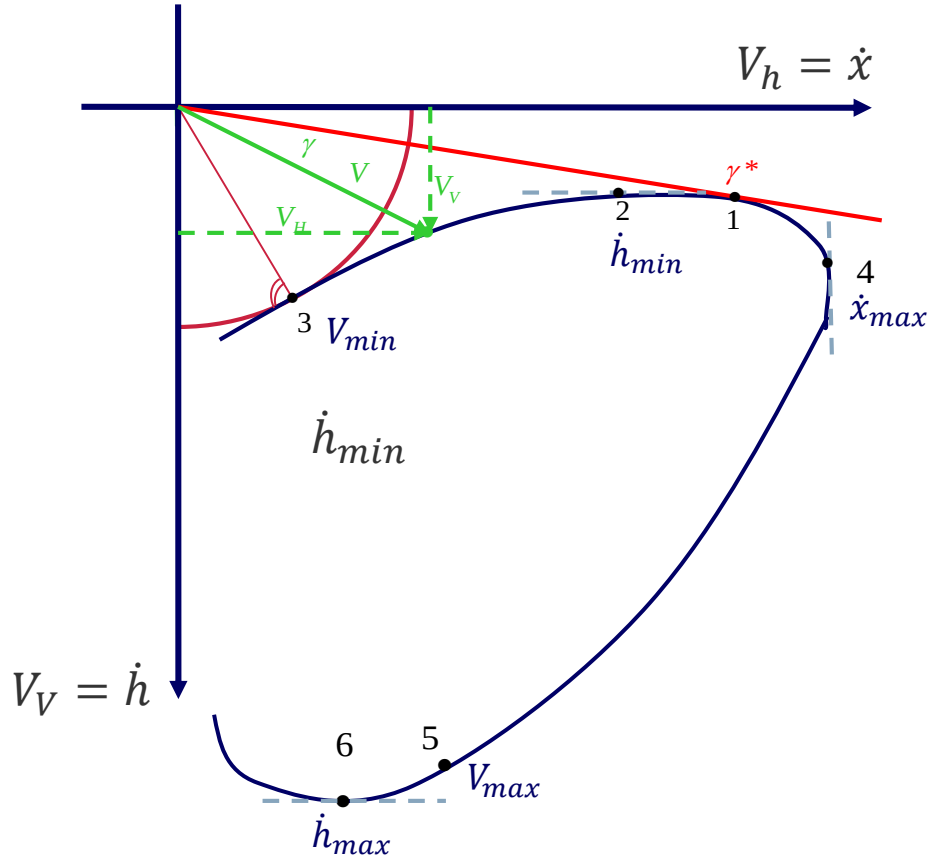
The ratio of both velocities is shown in the **„Velocity Polar“**.

For the glide angle it holds:

$$\frac{V_V}{V_H} = \frac{\dot{h}}{\dot{x}} = \frac{V \cdot \sin\gamma}{V \cdot \cos\gamma}$$

$$\frac{\Delta h}{\Delta x} = \tan\gamma$$

The glide angle is a measure for how much horizontal distance is covered after losing a specified amount of altitude.



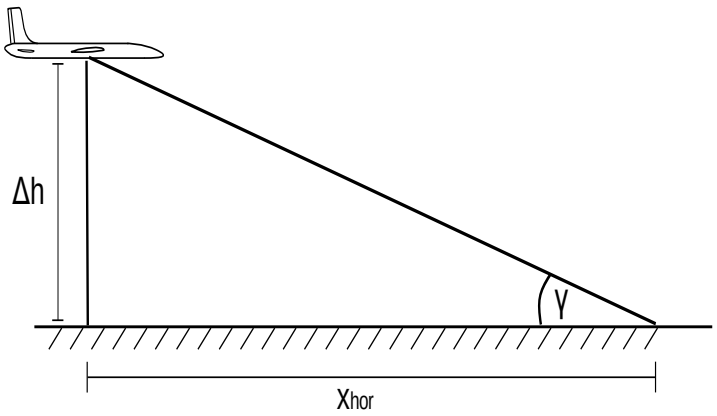
Note:
*The importance of the *-quantities goes far beyond gliding flight! For example, horizontal flight with velocity V* is drag-minimal. V* and ε* can also be used to set up equations for drag, velocity, or the climb angle. Since the *-quantities depend only on the aerodynamics of the aircraft, this results in very compact representations.*

$$C_L^* = \sqrt{\frac{C_{D0}}{k}}$$

$$C_D^* = 2C_{D0}$$

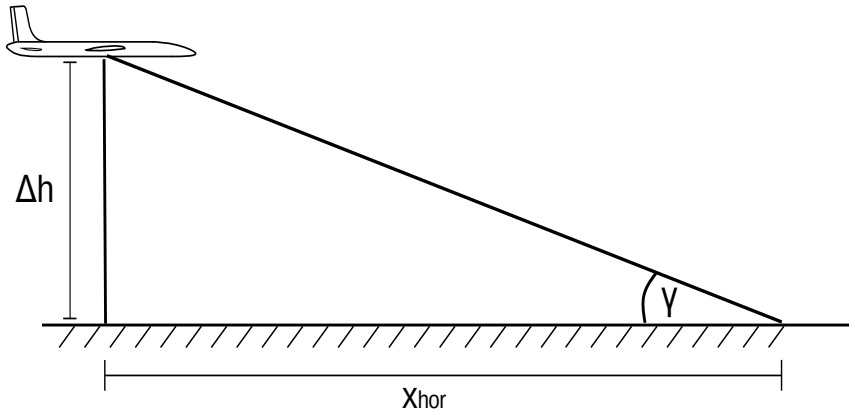
$$V^* = \sqrt{2\frac{m\,g\cdot 1}{S\,\rho\sqrt{\frac{C_{D0}}{k}}}}$$

$$\varepsilon^* = 2\sqrt{k\,C_{D0}}$$



$$\frac{\Delta h}{\Delta x} = \tan\gamma$$

What is the **maximum horizontal distance** the aircraft can cover?



From the lecture we know these two relations:

$$\frac{\Delta h}{\Delta x} = \tan\gamma \qquad \varepsilon = \frac{C_D}{C_L} = -\tan\gamma$$

In order to get Δx_{max}, we only have to insert ε^*:

$$\varepsilon^* = 2\sqrt{k\, C_{D0}}$$

$$\Delta x_{max} = \frac{\Delta h}{-\varepsilon^*} = \frac{-1000}{-2\sqrt{0.026 \cdot 0{,}031}}\, m = 17600\text{m}$$

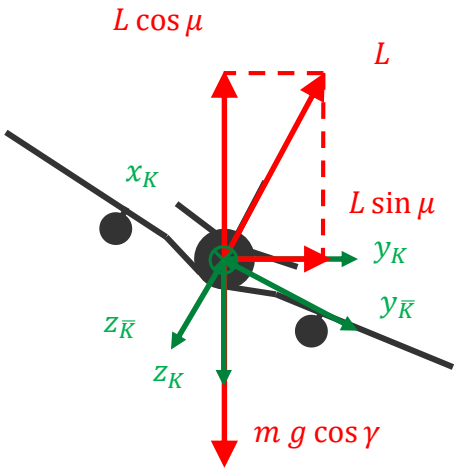
Exercise 5.2

What are the **control inputs of the point mass model** and **how can the pilot influence them**?

$$\dot{V} = \frac{T - D}{m} - g \sin(\gamma_K)$$

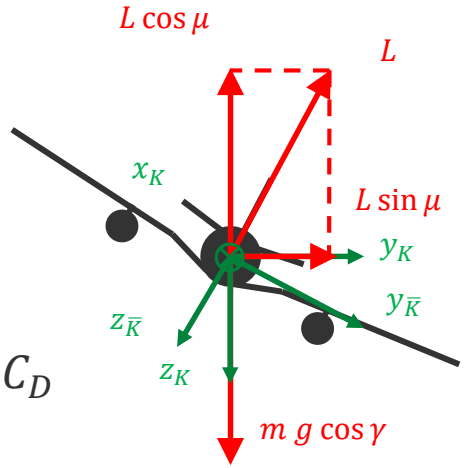
$$\dot{\chi} = \frac{L \sin \mu_K}{m\, V \cos(\gamma_K)}$$

$$\dot{\gamma} = \frac{L \cos \mu_K}{m\, V} - \frac{g}{V} \cos(\gamma_K)$$



- The states (V, χ, γ) depend on the forces (L, D, T, mg)
- Evidently, we cannot change mg ...
- However,(L, D, T) can be changed!

So how can we change the magnitude & direction of the forces (L, D, T)?



- **Direction** of L via (kin.) bank angle μ_K
- Usually the **direction** of (D, T) cannot be directly controlled by the pilot
- **Magnitude** of (L, D) via C_L: $L = \frac{1}{2}\rho V^2 S\, C_L, D = \frac{1}{2}\rho V^2 S\, C_D$
- , where C_D follows from drag polar: $C_D = C_{D0} + k\, C_L^2$

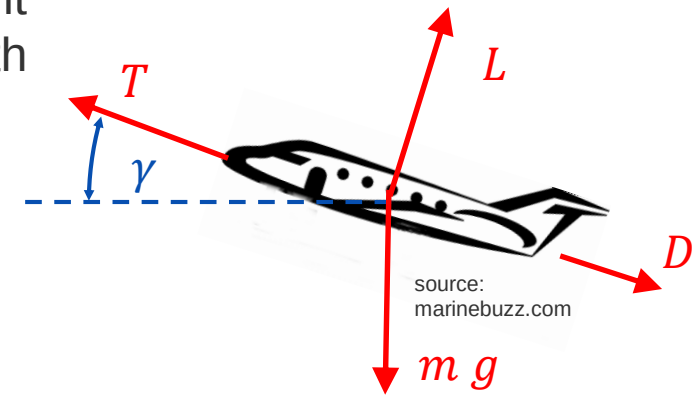
- **Magnitude** of T via thrust lever δ_T: $T = L_P \cdot V^{n_V} = \delta_T \cdot L_{P,max} V^{n_V}$

Control	Influence	Pilot
δ_T	Thrust force	Thrust lever
C_L	Magnitude of lift/drag	Control surfaces (stick)
μ_K	Direction of lift	Control surfaces (stick)

Control inputs to the PM-model

Definition Climb and Descent:

Pure climb and descent are defined as straight flight without maneuvers but constant flight path angle γ.



Straight flight:	$\dot{\chi} = 0$	
Stationary:	$\dot{\gamma} = 0$	$\dot{V} = 0$
Constant flight path:	$\gamma = \text{const.}$	

Important Performance Parameters for Climb and Descent:

1. *Which thrust is desired for climbing?* → $T \; bzw. \frac{T}{m \cdot g}$
2. *Climb anlge dependency on velocity?* → $\gamma \; (V)$
3. *Thrust reserve for maneuvers?*

Using the definition of climb and descent in the point mass equations of motion:

$$\dot{V} = \frac{T - D}{m} - g \sin\gamma = 0 \quad \longrightarrow T = mg \sin\gamma + D$$ ⇦ Stationary

$$\dot{\chi} = \frac{L \sin\mu}{m\, V \cos\gamma} = 0 \quad \longrightarrow \mu = 0$$ ⇦ Stationary and straight

$$\dot{\gamma} = \frac{L \cos\mu}{m\, V} - \frac{g}{V}\cos\gamma = 0 \quad \longrightarrow L = m\, g \cos\gamma$$ ⇦ Stationary, straight and without maneuvers

($\cos\mu = 1$)

$$\longrightarrow \tan\gamma = \frac{T - D}{L}$$

$$\longrightarrow \sin\gamma = \frac{T - D}{mg}$$

Goal: Deriving equation for thrust-to-weight ratio.

Calculation of Performance Parameters

Rearranging the equation yields:

$$\sin\gamma = \frac{T - D}{mg} \longrightarrow \frac{T}{mg} = \sin\gamma + \frac{D}{mg}$$

$$\frac{D}{L} = \frac{C_D}{C_L} = \varepsilon \quad \Longrightarrow \quad D = \varepsilon L = \varepsilon\, mg\cos\gamma$$

$$\frac{T}{mg} = \sin\gamma + \frac{\varepsilon\, mg\cos\gamma}{mg} \longrightarrow \boxed{\frac{T}{m\,g} = \sin\gamma + \varepsilon\cos\gamma}$$

The thrust-to-weight ratio solely depends on the flight path angle and the glide number.

Calculation of Performance Parameters

Like for the straight horizontal flight, the„*-quantities“ offer a more compact presentation:

$$\sin\gamma = \frac{T}{m\,g} - \frac{C_D}{C_L}\cos\gamma$$

Assumption: symmetric quadratic polar

$$\frac{T}{m\,g} - \left(\frac{C_{D0}}{C_L} + k\,C_L\right)\cos\gamma$$

$$\left(\frac{V}{V^*}\right)^2 = \frac{\frac{2\,m\,g}{\rho\,S\,C_L}\cos\gamma}{\frac{2\,m\,g}{\rho S\,C_L^*}} = \frac{C_L^*}{C_L}\cos\gamma \quad \rightarrow C_L = \left(\frac{V^*}{V}\right)^2 C_L^*\cos\gamma$$

$$\sin\gamma = \frac{T}{m\,g} - \frac{C_{D0}}{C_L^*}\left(\frac{V}{V^*}\right)^2 - k\,C_L^*\left(\frac{V^*}{V}\right)^2\cos^2\gamma$$

$$C_L^* = \sqrt{\frac{C_{D0}}{k}} \rightarrow k\,C_L^* = \frac{\varepsilon^*}{2}, \quad \varepsilon^* = \frac{C_D^*}{C_L^*} = \frac{2C_{D0}}{C_L^*} \rightarrow \frac{C_{D0}}{C_L^*} = \frac{\varepsilon^*}{2}$$

$$\sin\gamma = \frac{T}{m\,g} - \frac{\varepsilon^*}{2}\left(\frac{V}{V^*}\right)^2 - \frac{\varepsilon^*}{2}\left(\frac{V^*}{V}\right)^2\cos^2\gamma$$

Solving for $\frac{T}{m\,g}$

$$\frac{T}{m\,g} = \sin\gamma + \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\cos^2\gamma\right]$$

For small flight path angles γ, we can assume ($\cos^2\gamma \approx 1$) and solve for the flight path angle:,

$$\gamma(V) = \arcsin\left[\frac{T}{m\,g} - \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\right]\right]$$

Thrust-to-weight ratio

A derivation similar to the derivations for the straight horizontal flight shows, that that this quantity equals the ratio of drag and weight $\frac{D}{m\,g}$.

Summary: The climb angle results from the difference of thrust and drag. Higher thrust than drag leads to a positive climb angle, while higher drag leads to a negative climb angle.

The slides for gliding already contained the relations between change of altitude $\dot{h}$ and flight path angle γ:

$$V_V = \dot{h} = V \cdot \sin\gamma \xrightarrow{\sin\gamma = \frac{T}{m\,g} - \frac{D}{mg}} \boxed{\dot{h} = V \cdot \left(\underbrace{\frac{T}{m\,g}}_{spec.\,Thrust} - \underbrace{\frac{D}{mg}}_{spec.\,Drag} \right)}$$

General equation for power: $\boxed{P = F \cdot V}$

Equation for specific power: $\boxed{P_{spec} = \frac{F \cdot V}{mg}}$

The above equation for $\dot{h}$ equals the definition of a specific power with:

$$\dot{h} = \left(\underbrace{\frac{T \cdot V}{m\,g}}_{spec.\,propulsion\,power} - \underbrace{\frac{D \cdot V}{mg}}_{spec.\,power\,dissipated\,by\,drag} \right)$$

This power excess is also called **„specific excess power (SEP)“** and is a measure for the power available for climbing

c) Derive the **velocity which is required for reaching the maximum climb angle**.

In order to reach the max climb angle we need full thrust: $T = L_{P,max}$

From the lecture we know how the climb angle γ depends on the thrust T and the velocity V:

$$\gamma(V) = \arcsin\left[\frac{T}{m\,g} - \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\right]\right]$$

To find the maximum, we take the first derivative of $\gamma(V)$ w.r.t. V and set it to zero:

$$\frac{d\gamma}{dV} = 0$$

Hint:
For angles between. 0° und 90°, arcsin is maximal if its argument is maximal.
Thus, we maximize its argument directly:

$$f(V) = \frac{T}{m\,g} - \frac{\varepsilon^*}{2}\left[\left(\frac{V}{V^*}\right)^2 + \left(\frac{V^*}{V}\right)^2\right]$$

Does not depend on V!

$$\frac{d\gamma}{dV} = 0 \quad \longrightarrow \quad \frac{df}{dV} = 0$$

$$\frac{df}{dV} = -\frac{2\cdot\varepsilon^*}{2}\frac{V}{V^{*2}} + \frac{2\cdot\varepsilon^*}{2}\frac{V^{*2}}{V^3} = -\frac{V\cdot\varepsilon^*}{V^{*2}} + \frac{V^{*2}\cdot\varepsilon^*}{V^3} = 0$$

$$\frac{V}{V^{*2}} = \frac{V^{*2}}{V^3} \rightarrow V = V^*$$

Intuitive result, since V^* minimizes drag D for a given thrust T.
At this velocity there is the largest difference between T and D.
Here, we can achieve the largest climb angle γ.

… Which climb angle and which velocity is reached?

$$V^* = \sqrt{2\frac{2000\ 9{,}80665 \cdot 1}{18\ 1{,}1117\sqrt{\frac{0{,}031}{0{,}026}}}\frac{m}{s}} = 42{,}37\frac{m}{s}$$

$$\gamma(V^*) = \arcsin\left[\frac{T}{m\ g} - \frac{\varepsilon^*}{2}\left[(1)^2 + (1)^2\right]\right] = 0{,}252\ rad = \ 14.43°$$

Exercise 5.4

Now the aircraft is supposed to fly a **turn** with the data given below. Which **kinematic bank angle** and what **velocity** is required?

Height and density	$h \ / \ \rho$	$1000\ \mathrm{m} \ / \ 1.1117\ \frac{\mathrm{kg}}{\mathrm{m}^3}$
Turn rate	$\dot{\chi}$	$5\ \frac{^\circ}{\mathrm{s}}$
Lift coefficient turn	$C_{L,turn}$	1

Using the definition of turn in the point mass equations of motion

$$\dot{V} = \frac{T - D}{m} - g \underbrace{\sin\gamma}_{=0} = 0 \qquad \dot{\chi} = \frac{L \sin\mu}{m\, V \underbrace{\cos\gamma}_{=1}} \qquad \dot{\gamma} = \frac{L\cos\mu}{m\, V} - \frac{g}{V}\underbrace{\cos\gamma}_{=1} = 0$$

$$\rightarrow T = D \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\rightarrow L \sin\mu = mV\dot{\chi} \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\rightarrow L\cos\mu = m\, g \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\frac{L\sin\mu}{L\cos\mu} = \frac{m\, V\, \dot{\chi}}{m\, g} \rightarrow \tan\mu = \frac{V\,\dot{\chi}}{g}$$

$$L^2\left(\sin^2\mu + \cos^2\mu\right) = (m\, V\dot{\chi})^2 + (m\, g)^2 \rightarrow L = m\, g\sqrt{1 + \left(\frac{V\,\dot{\chi}}{g}\right)^2}$$

Question: Is there a parameter defining the turn performance of an aircraft?

Introduction of the loadfactor n_z

For defining turn performance, we introduce the load factor n_z:

$$(n_z)_K = \frac{L}{m\,g}$$

Acceleration due to lift

Normalization with respect to weight

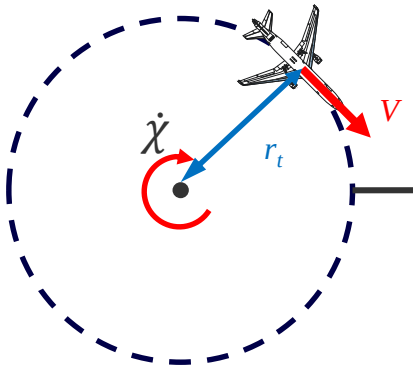
The load factor is the lift force normalized by the gravitational force. This normalization leads to $n_z = 1$ in straight horizontal flight.

$$\left(n_{z,Horizontal}\right)_K = 1$$

With this definition of the load factor, the available turn performance can be described, as the turn performance results from an excess of lift over weight.

Annotation: This interpretation of load factor is fit to the following turn performance. However, the benefit and usage of the load factor can be applied to more problems as well.

How can the turn radius be obtained?



The turn radius depends on aircraft velocity and turn rate

$$r_t = \frac{V}{\dot{\chi}}$$

Goal: dependency between turn rate, turn radius and n_z, in order to answer the question of minimum possible turn rate

$$n_z = \frac{L}{m\,g} \quad \xrightarrow{L = m\,g\sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2}} \quad n_z = \sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2} \quad \xrightarrow{\dot{\chi} = \frac{V}{r_t}} \quad n_z = \sqrt{1+\left(\frac{V^2}{g\,r_t}\right)^2}$$

$$n_z = \frac{1}{\cos\mu}$$

$$\tan\mu = \frac{V\,\dot{\chi}}{g} \quad \xrightarrow{\text{Solving for } \dot{\chi}} \quad \dot{\chi} = \frac{g}{V}\tan\mu \quad \xrightarrow{n_z = \sqrt{1+\tan^2\mu} \rightarrow \tan\mu = \sqrt{n_z^2-1}} \quad \dot{\chi} = \frac{g}{V}\sqrt{n_z^2-1}$$

Minimum Turn Rate

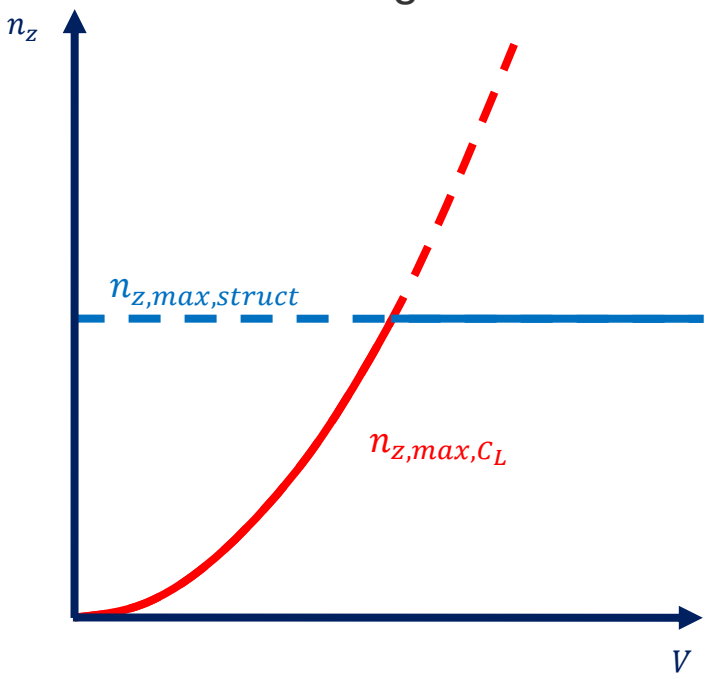
$$\dot{\chi} = \frac{g}{V}\sqrt{n_z^2 - 1} \qquad r_t = \frac{V}{\dot{\chi}} \qquad r_t = \frac{V^2}{g\sqrt{n_z^2 - 1}}$$

The turn radius is physically bounded by the maximum available load factor. A maximum load factor exists due to two reasons:

1. Both the pilot and the aircraft structure only withstand a maximum load factor.
2. The load factor is limited by the maximum achievable lift.

The minimum possible turn rate is defined by Lift, capability of structure and pilot to withstand loads, and maximum available thrust.

V-n Diagramm



Now the aircraft is supposed to fly a **turn** with the data given below. Which **kinematic bank angle** and what **velocity** is required?

From EOM.:

$$(1)\quad L\cos\mu = m\,g$$

$$(2)\quad L\sin\mu = mV\dot{\chi}$$

$$\longrightarrow \quad \tan\mu = \frac{V\,\dot{\chi}}{g} \qquad (3)$$

Aerodynamics

$$L = \frac{1}{2}\rho V^2 S\,C_L \qquad (4)$$

(4) and (1):

$$L = \frac{m\,g}{\cos\mu} = \frac{1}{2}\rho V^2 S\,C_L \qquad (5)$$

2 Equations for 2 Unknowns: μ, V

$$\tan\mu = \frac{V\,\dot{\chi}}{g} \quad \rightarrow \quad V = \frac{(g \cdot \tan\mu)}{\dot{\chi}} \rightarrow$$

Eq. (3) and (5):

$$\frac{m\,g}{\cos\mu} = \frac{1}{2}\rho\frac{g^2 \cdot \tan^2\mu}{\dot{\chi}^2} S\,C_L$$

Rearrange:

$$m \cdot g = \frac{\rho \cdot g^2 \cdot S \cdot C_L}{2 \cdot \dot{\chi}^2}\frac{\sin^2\mu}{cos\,\mu} \qquad (6)$$

With $\sin^2\mu + \cos^2\mu = 1$:

$$\underbrace{m \cdot g}_{t} = \underbrace{\frac{\rho \cdot g^2 \cdot S \cdot C_L}{2 \cdot \dot{\chi}^2}}_{s}\frac{\left(1 - cos^2\mu\right)}{cos\,\mu}$$

$$t = s \cdot \frac{(1 - \cos^2 \mu)}{\cos \mu} \rightarrow t \cdot \cos \mu = s - s \cdot \cos^2 \mu$$

Substitution with x: $s \cdot \underbrace{\cos^2 \mu}_{x^2} + t \cdot \underbrace{\cos \mu}_{x} - s = 0$

We end up with a quadratic formula:

$$x = \frac{-t \pm \sqrt{t^2 + 4s^2}}{2 \cdot s} \Rightarrow x_1 = -1.0802 \qquad x_2 = 0.9257$$

Considering that $x = \cos \mu$, only one of the two solutions is physically meaningful:

$$\mu = \text{acos}(0.9257) = 0{,}3887\ rad = 22{,}27\ °$$

$$V_{Turn} = \frac{(g \cdot \tan \mu)}{\dot{\chi}} = \frac{(9{,}80665 \cdot 0{,}4085)}{5 * \frac{\pi}{180}} \frac{m}{s} = 46{,}0 \frac{m}{s}$$